

Municipal Solid Waste Route Optimizer

Use-Case Flow Specification — Problem Statement #25

Use Case ID / Name:	UC-02 – Generate Optimized Route
Primary Actor:	Sanitation Supervisor
Supporting Actors:	Truck Driver (receives the resulting route)
Includes:	UC-01 – Monitor Bin Fill Levels (<<include>>)
Extended by:	UC-03 – Manually Tag Bin for Route (<<extend>>)

Preconditions

- The Sanitation Supervisor is authenticated and logged into the portal.
- Ultrasonic fill-level sensors are actively reporting bin data to the system (UC-01).
- At least one truck and driver are available and marked "on shift" for the day.

Postconditions

- Success: An optimized route manifest is created, saved, and pushed to the assigned Truck Driver's app (UC-04).
- Failure: No route is generated or saved, and the Supervisor is notified of the reason for failure.

Main Success Scenario

- 1 Sanitation Supervisor selects "Generate Route" for a chosen truck fleet and collection zone from the dashboard.
- 2 System executes UC-01 (Monitor Bin Fill Levels) to fetch the latest ultrasonic fill-level reading for every bin in the zone.
- 3 System filters bins, retaining only those with fill level $\geq 80\%$, plus any bins manually tagged by the Supervisor (UC-03, optional extension).
- 4 System runs the route-optimization algorithm across the filtered bin list and the available truck fleet.
- 5 System calculates the shortest-time collection sequence for each truck, respecting truck capacity and shift-time limits.
- 6 System displays the proposed route(s) to the Supervisor as an ordered stop list with estimated completion time.
- 7 Supervisor reviews and approves the proposed route.
- 8 System saves the approved route and pushes the manifest to the corresponding Truck Driver's mobile app.
- 9 Use case ends successfully.

Alternate Flow

3a. No Bins Meet the Threshold

- 3a1. System finds no bins at or above 80% fill and no manually tagged bins in the selected zone.
- 3a2. System displays a "no collection required" message to the Supervisor.
- 3a3. Use case ends without generating a route.

6a. Optimization Exceeds Time Limit

- 6a1. Route calculation does not complete within the 45-second performance target (NFR-001).
- 6a2. System cancels the calculation and displays a timeout error to the Supervisor.

- 6a3. Supervisor may retry with a smaller zone/fleet selection or escalate to IT support.